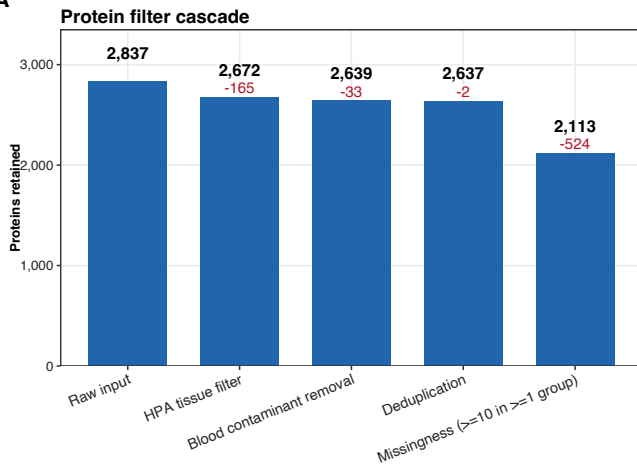


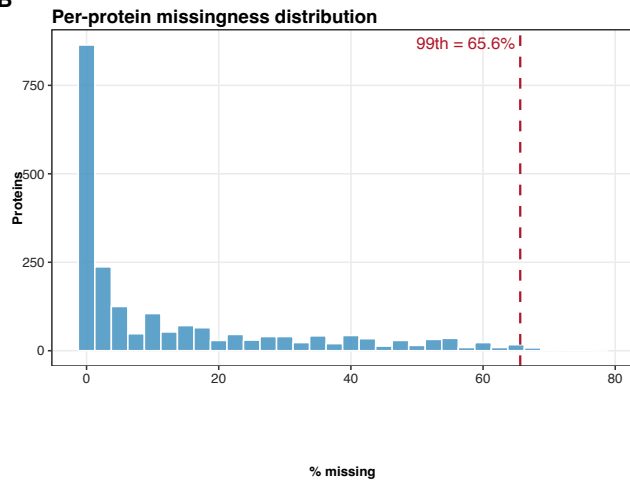
# Pipeline QC — Pre-Processing

DIA-NN → HPA filter → cyclic loess → outlier removal | 2,837 → 2,113 proteins × 62 samples

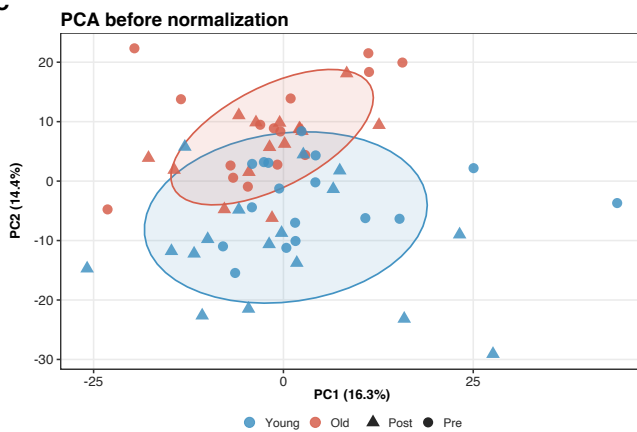
**A**



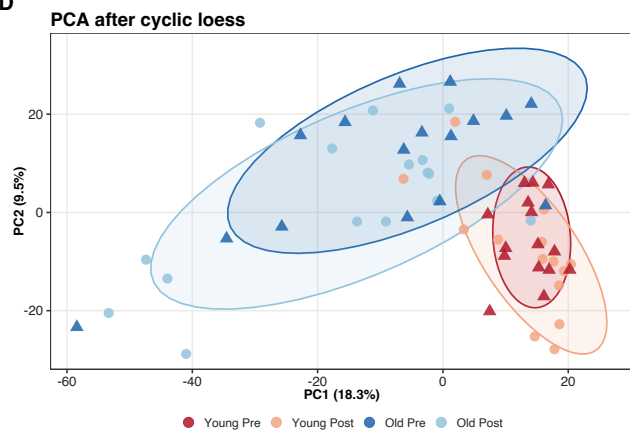
**B**



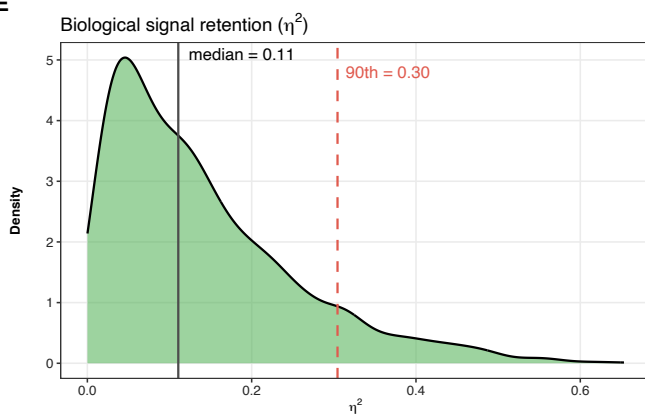
**C**



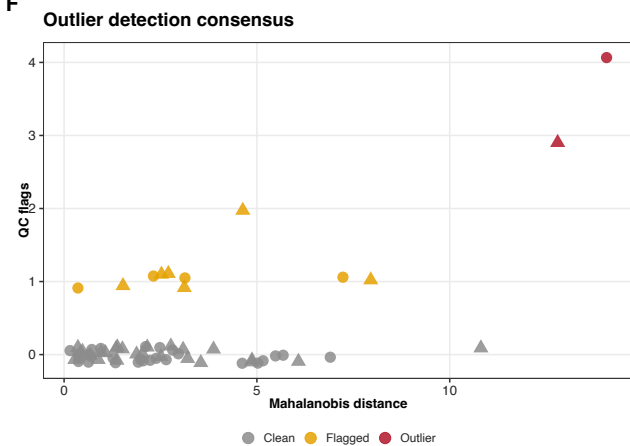
**D**



**E**



**F**



**G**

